Curriculum vitae

PERSONAL INFORMATION ---

STAVROS NOUSIAS

Professional Address: ARCISSTRASSE 21, MUNICH, 80333, GERMANY

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LinkedIn: https://www.linkedin.com/in/stavrosnousias/

Google Scholar: https://scholar.google.gr/citations?&user=HQuAxd0AAAAJ

EDUCATION & TRAINING ---

Doctor of Philosophy July 2022

Department of Electrical and Computer Engineering, University of Patras, Greece

Title: "Patient-specific modelling, simulation, and real time processing for respiratory diseases", 04-07-2022

Link: https://arxiv.org/abs/2207.01082 , http://hdl.handle.net/10889/16427

Master of Science in Electronics and Information Processing June 2016

Interdepartmental Master's Program of the University of Patras

Diploma in Electrical and Computer Engineering (Dipl.-Ing) (5-year curriculum) July 2011

Department of Electrical and Computer Engineering, University of Patras, Greece

Specialization: Computer Engineering & Informatics

PROFESSIONAL EXPERIENCE ---

Postdoctoral research scientist and group leader "Artificial Intelligence in Construction" Nov 2022 – Present

Chair of Computational Modeling and Simulation, Technical University of Munich, Arcisstrasse 21, 80333 München, Germany

Projects in the following areas: Robotic Construction, Pedestrian Dynamics, AI in Compliance Checking, AI in CAD and CAM, Analysis of Technical Drawings

Research associate / Research scientist Jan 2020 – Nov 2022

"Athena" Research and Innovation Center, Marousi, Attiki, 15125, Greece,

Field: Automotive Scene Analysis, Geometry, Point Cloud Processing, Heritage Science, Machine Learning,

Projects: CPSOSAWARE (H2020 Grant agreement: 871738), WARMEST (H2020-Marie Skłodowska-Curie: 777981)

Research Associate Jan 2019- Dec 2019

Visualization and Virtual Reality Lab, University of Patras Department of Electrical and Computer Engineering

Field: Machine learning applications in Healthcare, Research design, Prototyping, Integration, Technical writing.

Projects: GamECAR (H2020 Grant agreement: 732068), Ageing@Work (H2020 Grant agreement: 826299)

Research Engineer Jan 2015-Dec 2018

Visualization and Virtual Reality Lab, University of Patras Department of Electrical and Computer Engineering

Field: 3D processing, Prototyping, Integration, DevOps

Projects: MyAirCoach (H2020 Grant agreement: 643607)

PROFESSIONAL SERVICE --

Reviewer for IEEE ACCESS, Advanced Engineering Informatics, Elsevier Measurement Journal

RESEARCH INTERESTS --

Geometry processing, geometric deep learning, deep learning on 3D surface meshes, graph neural networks

TEACHING --

Artificial Intelligence in Engineering (Master Programme @ TUM)
Computation in Engineering I (Master Programme @ TUM)

LANGUAGES --

Greek: Native language, English: C2 Proficiency, German: Intermediate

SHORT LIST OF SELECTED PUBLICATIONS --

Doctoral dissertation

[D] Nousias, S., 2022. Patient-specific modelling, simulation and real time processing for respiratory diseases, Doctoral dissertation, University of Patras. Polytechnical School. Department of Electrical and Computer Engineering. http://dx.doi.org/10.12681/eadd/52290

Generative geometry models in the built environment

[J] Du, C., Esser, S., Nousias, S. and Borrmann, A., 2024. Text2BIM: Generating Building Models Using a Large Language Model-based Multi-Agent Framework. arXiv preprint arXiv:2408.08054.

Geometric deep learning

[J] Nousias, S., Arvanitis, G., Lalos, A. and Moustakas, K., 2022. Deep Saliency Mapping for 3D Meshes and Applications. ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM). https://doi.org/10.1145/3550073

[J] Nousias, S., Arvanitis, G., Lalos, A.S. and Moustakas, K., 2020. Fast mesh denoising with data driven normal filtering using deep variational autoencoders. IEEE Transactions on Industrial Informatics, 17(2), pp.980-990. https://doi.org/10.1109/TII.2020.3000491

[J] Nousias, S., Arvanitis, G., Lalos, A.S., Pavlidis, G., Koulamas, C., Kalogeras, A. and Moustakas, K., 2020. A saliency aware CNN-Based 3D Model simplification and compression framework for remote inspection of heritage sites. IEEE Access, 8, pp.169982-170001. https://doi.org/10.1109/ACCESS.2020.3023167

Computational and geometry modelling of the respiratory system

[J] Nousias, S., Zacharaki, E.I. and Moustakas, K., 2020. AVATREE: An open-source computational modelling framework modelling Anatomically Valid Airway TREE conformations. PloS one, 15(4), p.e0230259.

[J] Lalas, A., Nousias, S., Kikidis, D., Lalos, A., Arvanitis, G., Sougles, C., Moustakas, K., Votis, K., Verbanck, S., Usmani, O. and Tzovaras, D., 2017. Substance deposition assessment in obstructed pulmonary system through numerical characterization of airflow and inhaled particles attributes. BMC medical informatics and decision making, 17(3), pp.25-44. PMID: 29297393; PMCID: PMC5751792. https://doi.org/10.1186/s12911-017-0561-y .

Machine Learning for monitoring medication adherence in respiratory conditions

[J] Nousias, S., Lalos, A.S., Arvanitis, G., Moustakas, K., Tsirelis, T., Kikidis, D., Votis, K. and Tzovaras, D., 2018. An mHealth system for monitoring medication adherence in obstructive respiratory diseases using content-based audio classification. IEEE Access, 6, pp.11871-11882.

[J] Fakotakis, D.N., Nousias, S., Arvanitis, G., Zacharaki, E.I. and Moustakas, K., 2023. AI sound recognition on asthma medication adherence: Evaluation with the RDA benchmark suite. *IEEE Access*, *11*, pp.13810-13829.